

Grain Flow App Summary

Evidence source: README.md, docs/ARCHITECTURE.md, package.json, src/utils/supabase/middleware.ts, supabase/functions/*

What It Is

Grain Flow is a multi-tenant warehouse management web app for agricultural storage operations. It digitizes stock, customer, billing, payment, and subscription workflows using a Next.js + Supabase stack.

Who It Is For

ÿÿ Primary persona: warehouse owners/admins and operations staff managing grain storage, inflow/outflow, rent billing, and collections.

What It Does

- ÿÿ Tracks inflow, outflow, storage records, lot occupancy, and customer history.
- ÿÿ Calculates storage rent and tracks rent vs hamali (labor) payments.
- ÿÿ Provides dashboard metrics and operational reporting views.
- ÿÿ Supports role-based access and multi-warehouse membership/workflow.
- ÿÿ Offers customer/portal-style access patterns and profile-linked data views.
- ÿÿ Runs subscription lifecycle flows (renewal links, expiry checks, notifications).

How It Works (Repo-Evidenced Architecture)

- ÿÿ Frontend: Next.js App Router (`src/app`) with Server Components and client UI components (`src/components`).
- ÿÿ Auth + data: Supabase SSR clients (`src/utils/supabase/server.ts`, `client.ts`) and route middleware guard (`middleware.ts`) control access.
- ÿÿ Business logic: server actions and query modules in `src/lib/actions`, `src/lib`, and `src/lib/queries`.
- ÿÿ Database: Supabase PostgreSQL with migrations/RLS policies in `supabase/migrations` and docs in `docs/DATABASE.md`.
- ÿÿ Background services: Supabase Edge Functions and cron SQL (`supabase/functions/*`, `supabase/cron_setup.sql`) process subscription expiry/renewal tasks.
- ÿÿ Observability: Sentry Next.js configs (`sentry.client.config.ts`, `sentry.server.config.ts`, `sentry.edge.config.ts`).
- ÿÿ Message broker / event bus layer: Not found in repo.

How To Run (Minimal)

- ÿÿ Prereqs: Node.js 18+ and a Supabase project.
- ÿÿ Install deps: run `npm install` in repo root.
- ÿÿ Create `.env.local` with `NEXT_PUBLIC_SUPABASE_URL` and `NEXT_PUBLIC_SUPABASE_ANON_KEY`.
- ÿÿ Start dev server: `npm run dev` (package script runs on port 9002).